On the Power System of the AMALIA Moon Rover

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Abstract— **This work deals with the Electrical Power System of the Lunar Rover, developed for the AMALIA mission, intended to compete in the Google Lunar X prize. The main subsystems are described and their operation analyzed on the basis of the observed experimental behavior. Primary generation is provided by triple-junction solar cells, arranged as four sections, each connected to a dedicated Array Power Regulator with separate Maximum Power Point Tracking. A Switch Matrix allows to rearrange the inter-connections among the Solar Array sections and the APR in order to manage possible failures. During load peaks and/or with low Sun elevation the solar array is supported by a secondary generator consisting of three redundant units with Li-Ion battery cells. Two alternate configurations of the Battery Subsystem were developed: a classical 8S Li-Ion battery, for a 28V Sun-regulated bus, and a novel approach, based on a single-cell (1S) Li-Ion battery and a high-voltage-gain Battery Charge-Discharge Regulator. A specific converter was designed for this configuration, based on the non-insulated Three Levels Neutral Point Clamped switching converter topology. Its correct operation with bus voltages up to 48V and fully discharged battery cells was demonstrated.**

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1. INTRODUCTION

The Electrical Power System (EPS) of a rover is presented. It was developed by the Team Italia as a part of the AMALIA mission, aimed to reach and explore the Moon surface by December 2016 in the framework of the Google Lunar X - prize.

The AMALIA rover is equipped with a four-wheels locomotion system and an optical guidance system. It can communicate with an Earth Station either via a low-data-rate Direct-to-Earth link or via an high-data-rate Proximity Link using the lander as a relay. Both antennas and cameras are placed on a mast, to enhance their range (Figure 1) [1]. Mission-specific requirements for the EPS arise from the unforeseeable power profile of the locomotion subsystem, related to the random distribution of surface slopes and obstacles. In case of landing in a nearly equatorial site, only short eclipses are expected: they may occur when the Rover is escaping away from the Lander or, less likely, when it is passing through the shade of a boulder, like those encountered during the Apollo 17 mission. Nevertheless, secondary power from battery may be required during the power-demanding Direct To Earth transmissions. In case of mission control via a single Earth Antenna, long standby times would be used for battery charging.

The lunar environment [2] mainly affects thermal design. The scheduled duration of surface activities is shorter than a lunar day, corresponding to an overall mission duration shorter than 300 hours (also including flight). This allowed to relax requirements about total dose effects Thus the qualification level of most components was set to standard military, and just a few Commercial-Of-The-Shelf (COTS) components were kept, but only in highly redundant circuits. No rad-hard components were used. On the other hand, Single-Event-Effects cannot be excluded and were managed by design.

The circuit design was derived from that of a Power Conversion and Distribution Unit (PCDU) previously flown in a microsatellite, launched in LEO orbit in 2012. Most modifications were intended to improve reliability of specific circuits and functions. Two alternate designs were developed for the Battery Charge Discharge Regulator (BCDR): a classic configuration for supplying a sun-regulated 28V bus from an 8S array of Li-Ion cells and a novel design for a regulated bus at the same voltage, supplied by a single (1S) high-energy Li-Ion cell. This solution, suggested by availability of battery cells capable to provide the full capacity required for the mission, was accomplished after a high-gain BCDR was specifically developed.

In the following sections, after introducing the overall set-up of the rover, analyses and experiments on the single units of power system are presented. A section is dedicated to the BCDR for the single-cell batteries and to the experimental results achieved by the related demonstrator.

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2. SYSTEM CONFIGURATION

During daylight time the EPS is supplied by a triple junction photovoltaic primary source and uses a Li-Ion secondary source mostly to face load peaks. During the lunar night the Rover is allowed to freeze and no device for temperature control or Radioisotope Heating Unit [3] is provided.

Reliability issues mainly arise from mechanical stresses during launch and deployment: as a matter of fact wear out failures or degradation by radiation dose are not an issue due to the short mission duration. Suitable redundancies have been introduced at the subsystem level.

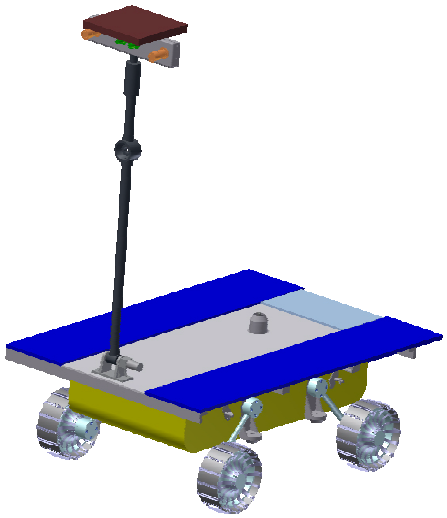


Figure 1. Artist view of the AMALIA Lunar Rover.

Solar Array

The Solar Array (SA) was designed to satisfy power demand in nominal mode with a worst-case Sun elevation of 45°. Otherwise, the maximum load can be supplied only with Sun elevation close to 90°. Hence in this environment partial failures of the SA are allowed and a 3/4 redundancy is achieved (Figure 2), as they just would cause a reduction of the operational time in high-power modes and an extension of the battery recharging time. With lower irradiation levels during power peaks it is necessary to draw secondary power from the battery Sub-System (S/S).

The SA shares the upper surface of the warm box with heat radiators. The allocated space for SA consisted of 2 areas, each organized as two strings. (see Figure 3).

The timelines in Figure 4 report the battery State Of Charge (SOC) and the available power from SA (P_{PV}) as a function of Sun elevation for a case-study mission profile tailored on a single Earth Antenna, with a standby time during every terrestrial day and maximum 12 hours of daily operation. The power absorption was considered a constant, at the maximum load.

Normally, the Rover is deployed after the power generated from SA exceeds the standby power. Nevertheless, using the stored energy it would be possible to anticipate the start of operations: in the case-study mission considered in Figure 4 the Rover was deployed when the Sun elevation was about 20°, resulting in full battery discharge after nearly 4 hours. In the subsequent stand-by time, the available energy from SA was not able to completely recharge the battery. As a consequence, even the subsequent operating time would have a reduced duration. Only with Sun elevation exceeding 45° the available power allowed to reach the target of 7 operating hours in 24h. The residual SOC after each operating slot increased with Sun elevation.

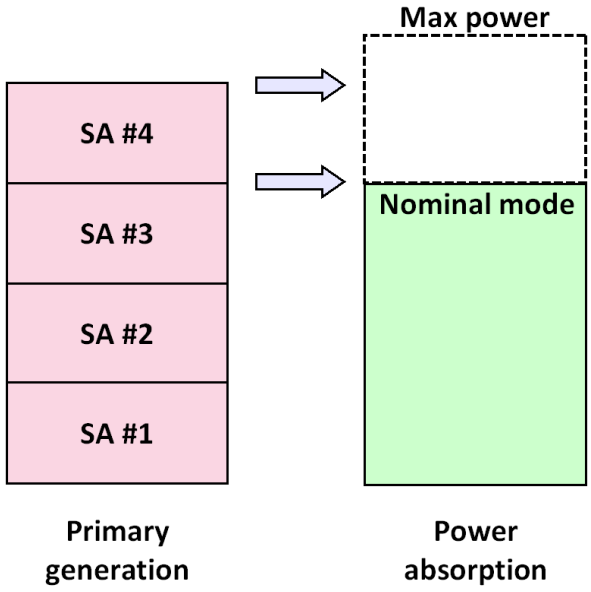


Figure 2. Available power from Solar Array with 45° Sun elevation.

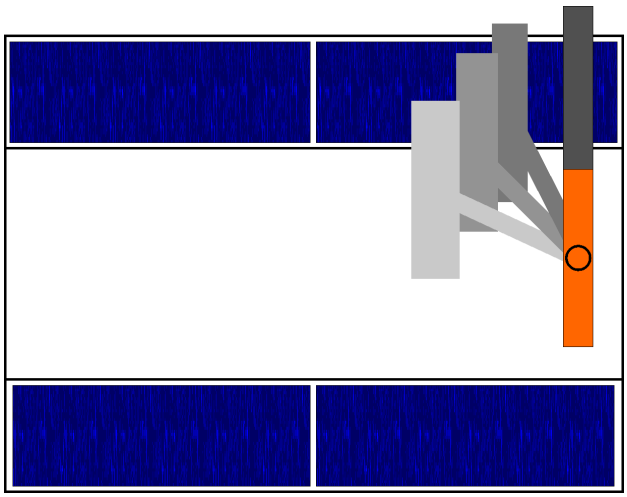


Figure 3. Outline of the solar panel and the shadowing effect caused by the rover mast.

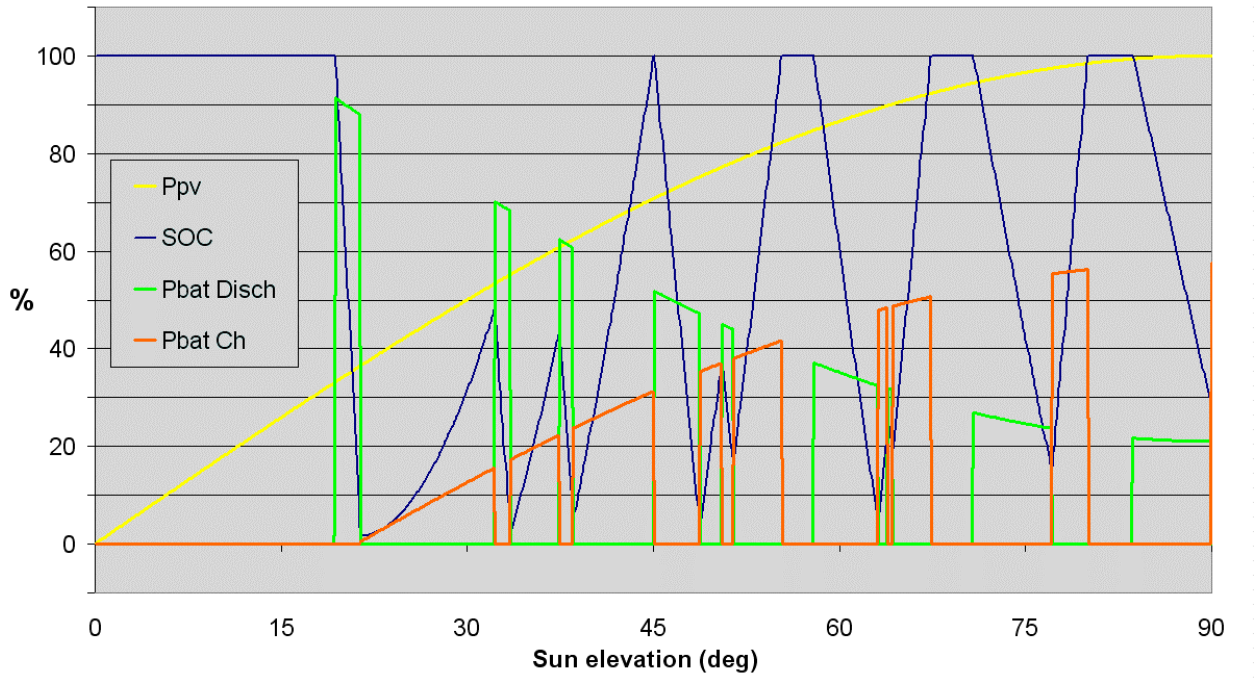


Figure 4. Battery capacity, SA available power and rover operating time during the mission.

In the terrestrial days between the 5th and the 9th after the local sunrise the rover was able to provide 12 hours continuous operation at maximum power.

Power Converting Unit

The Power Converting Unit (PCU) consists of a Switch Matrix and 4 Array Power Regulators (APR) (Fig. 5), each connected to a single string of the SA (at Begin Of Life).

This approach was preferred to other configurations based on the Sequential Switching Shunt Regulation (S3R) or on parallel converters because it can manage more effectively the string unbalances caused by shading or by single cell failures. Actually, cell shading by the mast is very frequent with low Sun irradiation (Figure 3): when it happens the string voltage can be reduced by several volts so that, in a low-voltage system with parallel strings, complete loss of one string could happen. In a S3R system, as the bus voltage

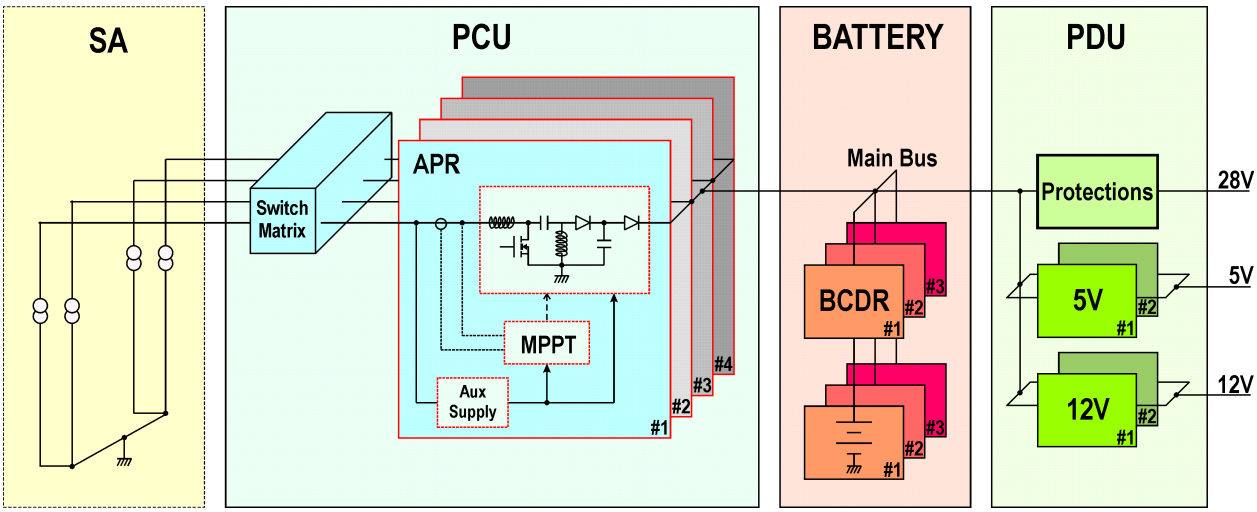


Figure 5. Block diagram of the SA and PCDU for the AMALIA Rover.

is linked to the SA voltage, the effects of shading could be only faced by increasing the string length with respect to the minimum required for the nominal voltage. Otherwise, the selected configuration allows to set optimal operation points for any string independently of the string length as it avoids string paralleling and provides independent Maximum Power Point Tracking (MPPT) controllers.

The Switch Matrix provides graceful degradation of the system, allowing to extend its life in case of failures in APR's. Its operation is sketched in Figure 6: in the presence of one failed APR the current generated from the corresponding SA string is readdressed by means of a network of switches, which are automatically managed on board or can be commanded by Earth. The adopted configuration for the Switch Matrix allows to connect a single APR to up to four strings, enabling to manage up to 3 failures [4]. On the other hand in our prototype the redundancy scheme could be considered to be only 2/4 as the design margin used in APR design does not allow a single regulator to manage the maximum power from the whole SA. Moreover, after paralleling the advantage of dedicated MPPT is lost and string mismatches can cause some power losses.

For the sake of reliability, the switches have been designed Single-Point-Failure (SPF) free. This task is accomplished by the circuit reported in Figure 7, consisting in a network of 6 MOSFETs. This configuration provides two alternate current paths for each way, allowing an increased reliability against switch clipping in either position. Robustness against double open circuit is also slightly increased.

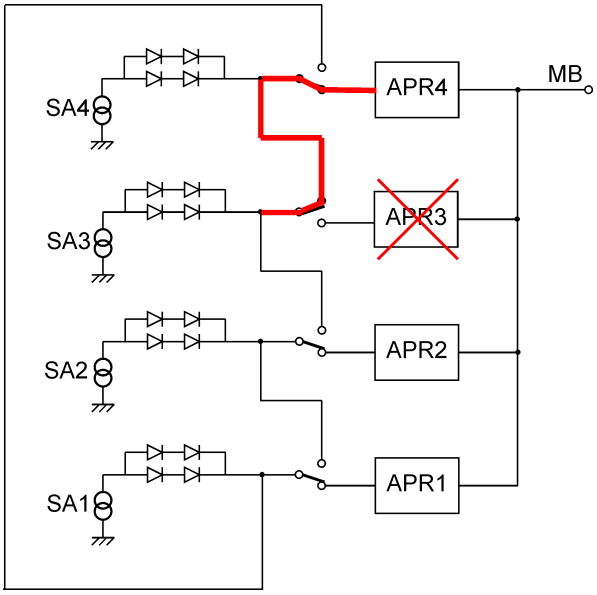


Figure 6. The switch matrix operation: after a failure in APR3 the SA3 output is redirected to APR4.

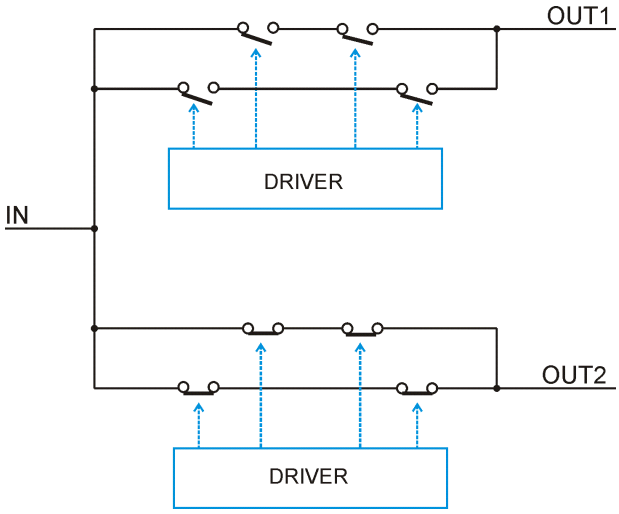


Figure 7. The configuration of the SPF free switch.

The APRs are based on the SEPIC (Single Ended Primary Inductor Converter) configuration outlined in Figure 8. This kind of converter was selected as it can provide a fixed output voltage from any input voltage (both lower and higher than output), allowing to relax specifications about the solar array voltage. Its circuit complexity is not larger than for a classical superbuck converter (Figure 9) and in case of failure it may take advantage from additional insulation provided by the series capacitor.

The PCU design benefits from the experience gained with a unit previously flown in a Low Earth Orbit (LEO) microsatellite [4].

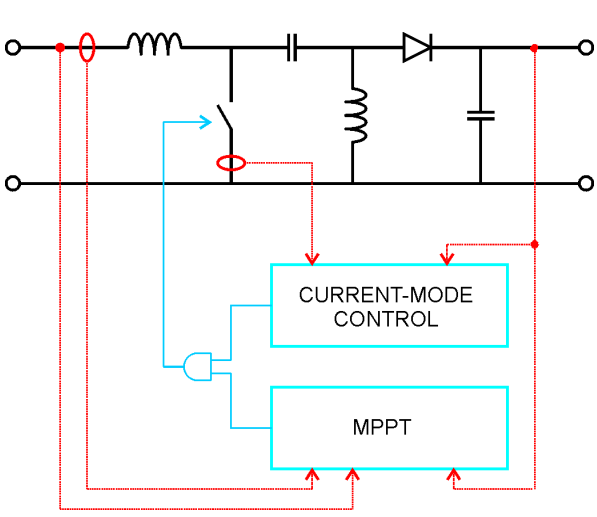


Figure 8. SEPIC converter with analog current-mode control and digital MPPT.

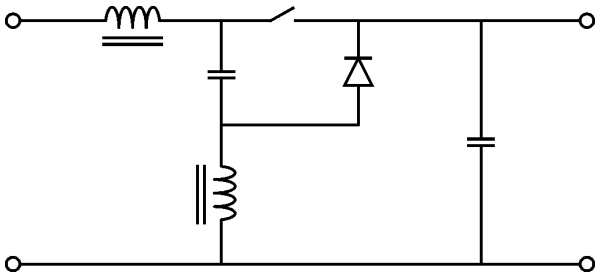


Figure 9. The Superbuck converter.

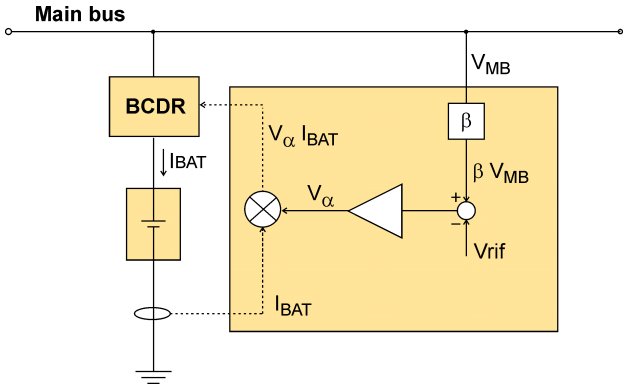


Figure 10. A schematic diagram of the MB and battery charging management.

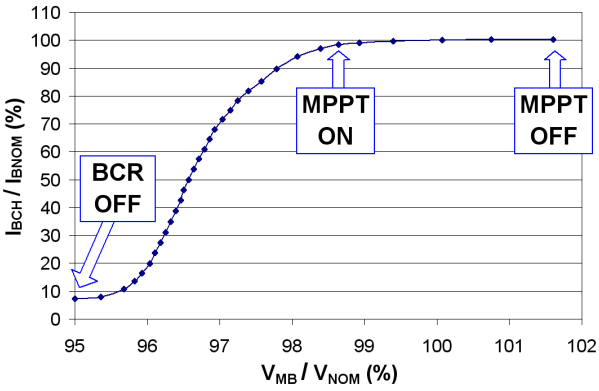


Figure 11. Battery charging current Vs. MB voltage.

The design was upgraded to improve heat dissipation and increase overall reliability. The COTS components were partially replaced with the corresponding Military Qualified components. Interfaces for the remaining COTS components were redesigned in order to minimize the effects of possible failures.

For example, duty cycle generation is provided by a redundant controller (see Figure 8) where a basic discrete-components circuit for duty-cycle generation [5] is normally overridden by a digital control implementing the SMP technique for MPPT management [6]. The controller is a COTS device whose correct operation is continuously checked so that it can be automatically disabled as soon as a malfunction is detected.

Bus and Power Distribution

The Main Bus (MB) is kept at a 28V nominal voltage. It is distributed to several payloads and subsystems by a switch board also providing monitoring and protection to single lines. Additional payload buses at 5V and 12V are provided, with an overall power smaller than 10% of total rover absorption: each of them is supplied by a pair of buck converters in hot redundancy (Figure 5).

In Battery Charging mode the MB voltage is controlled by the feedback system sketched in Figure 10, implementing a kind of linear shunt regulation. In this operational mode the APRs are normally operating in MPPT and behave as independent current sources, and the battery current is modulated in order to oppose to variations of the bus voltage associated to fluctuations of the excess power provided by the SA.

The resulting trend of the charging current is reported in Figure 11: it is possible to observe that small variations of the MB voltage are allowed. They can be considered as an error signal in the feedback control setting the battery current.

Two solutions were developed for bus control during Battery Discharge (Figure 12): full regulation or sun regulation, depending on the selected battery voltage range. They will be analyzed in the next section.

Battery subsystem

The Battery S/S comprises three units in redundancy 2/3. It is mainly intended to support primary generation during load peaks. In fact, the SA is capable of supplying all nominal power for high Sun elevations and only infrequent and short eclipses are expected.

A design based on the classical configuration with a 8S Li-Ion battery was developed (Figure 12a). The battery is manufactured by ABSL and consists of an array of Sony 18650 cells. This approach is interesting for its flexibility, given by the small capacitance of single cells, but it gives some concerns about reliability, mainly due to the large number of components. Moreover, correct operation is specified for a relatively small temperature range.

An alternative design was developed for increased reliability and reduced mass (Figure 12b). Each battery unit consists of a single high-energy cell, like the VES cells manufactured by Saft, and is interfaced with the MB by a high-gain converter.

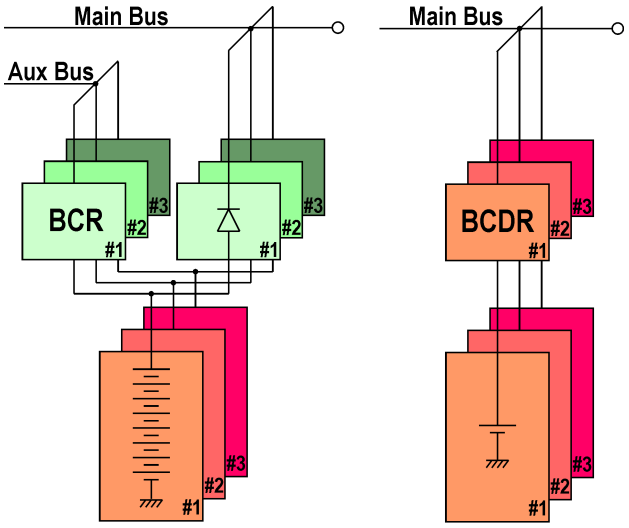


Figure 12. Different design approaches for the Battery S/S: (a) sun regulated with 8S batteries; (b) fully regulated with single-cell batteries.

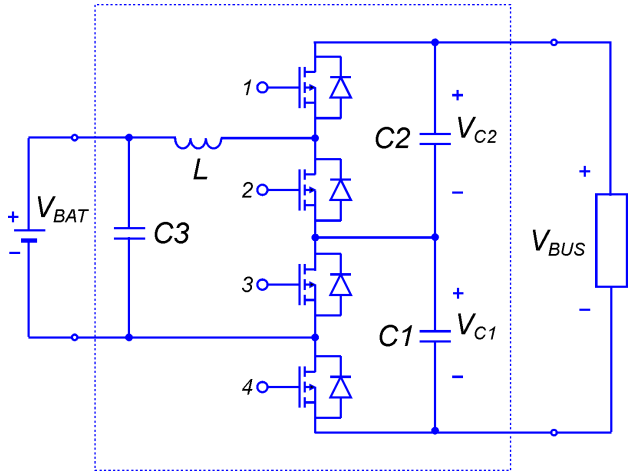


Figure 13. Schematic of the TLNPC bidirectional converter

This solution would allow to cut down the battery harness and to get rid of cell mismatching management [7] [8]. It also provides large flexibility in the choice of the MB voltage. It was enabled by several factors: first by the availability of high capacity battery cells from Saft, capable to meet the requirements about battery capacity by a single cell. Then the power of the AMALIA rover (in the range of some tens of W) was such as to keep the peak current within the safe operation range of available transistors.

Lastly, a converter with the suitable voltage gain and an acceptable conversion efficiency was specifically developed (Section 3). As an additional benefit, the converter acting as a Battery Discharge Regulator (BDR) provides bus voltage regulation, simplifying the payload interfaces with respect to Sun regulated approaches.

3. The High Gain Converter

A non-insulated high-gain converter was developed. The selected configuration was the synchronous Three Level Neutral Point Clamped (TLNPC) converter reported in Figure 13, well known for its inherent capability to provide high voltage gain with low stress to components [9].

For the sake of part count reduction we chose to use it as a bidirectional converter to operate as a BCDR. On the other hand, a different approach consisting of a high-current boost BDR based on the TLNPC topology and a small-power Battery Charge Regulator (BCR) would be more efficient, at the expense of an increased part count and volume.

A prototype was built and experimentally demonstrated the possibility to supply a bus at any voltage in the range 12-48V by a single Li-Ion cell. The discharge was tested up to 30A, with a battery voltage in the range 4.2-3.0V [10].

When the bus voltage was at 28V and the current in the 10-20A range the observed efficiency was 94% for battery voltage at 4V and dropped to 92% when battery was discharged to 2.8V. Slightly lower efficiencies were observed when the output voltage was set at 48V (Figure 14). These efficiency levels are not far from those typical of the classic configuration with Sun regulated bus, if the voltage drop on blocking diodes and the increased absorption in wide-input-range dc-dc converters for payloads are accounted for.

These results were enabled by a "lossless" design of the converter board, including optimal sizing of power MOSFETs, careful design of switching sequences and low-reactance paths for pulsed currents [11].

In fact the maximum achievable gain G_V in the BDR (boost) mode is limited by power dissipation: it can be related to the duty-cycle D and to the power losses P_{LOSS} by the equation:

$$G_V = \frac{1}{1 - D + \rho} \tag{1}$$

A first-order expression of ρ is:

$$\rho = \frac{P_{LOSS}}{R_L I_B^2 (1 - D)} \tag{2}$$

where R_L is the load resistance and I_B is the battery current. As the converter is intended to operate as a BCDR, a fast and smooth transition between charging and discharging modes was also provided.

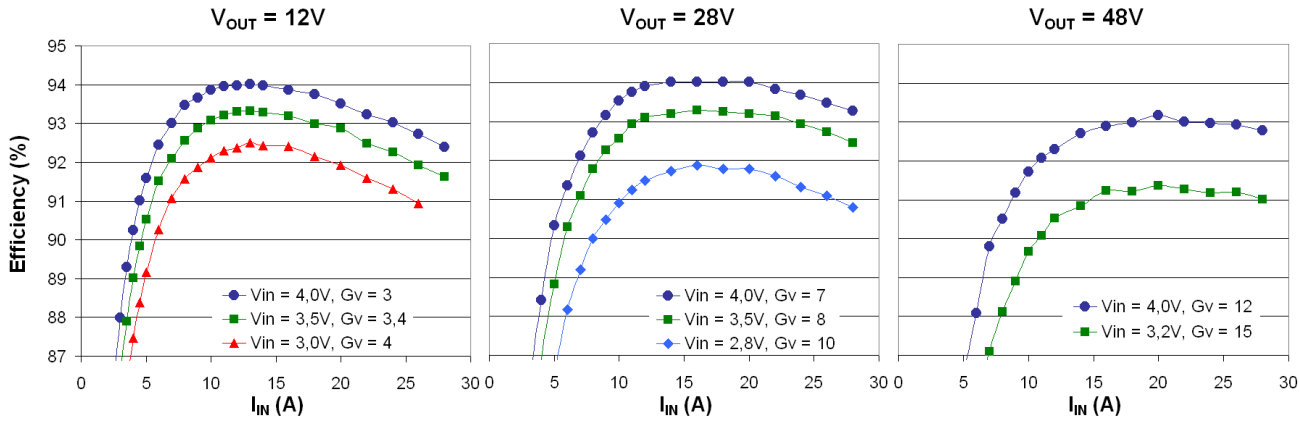


Figure 14: Overall efficiency Vs. input current at different bus levels with converter operating in BDR-boost mode with input voltages corresponding to a typical Li-Ion cell voltage range. The power absorbed by control circuitry is included in efficiency computations.

This feature was tested by applying step transitions to the load current while the BCDR was operating, so that the waveforms shown in Figure 15 were achieved.

It is possible to observe that initially the load current was low enough to allow excess power from Solar Array to charge the battery. After a positive load step of 33A the transient power deficit causes a voltage drop and the system switches from BCR at 5A to BDR at 28A. The voltage sag is nearly 2 V and the new stationary condition is reached in less than 30 ms.

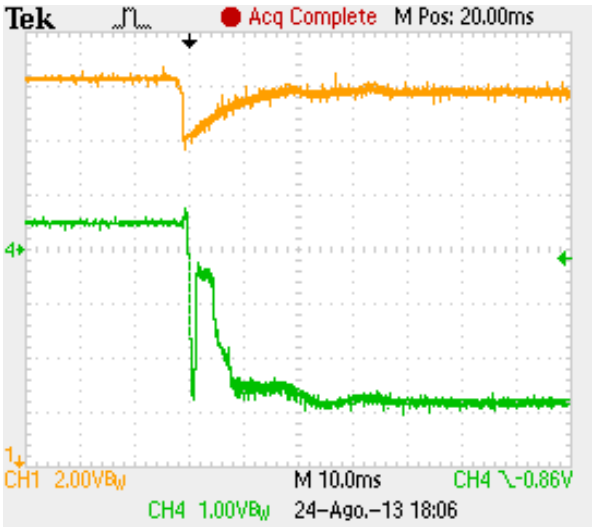


Figure 15. MB voltage and BCDR current during a step transition of the load current: the converter switches from BCR mode (5A charging current) to BDR mode (28A discharging current).

4. CONCLUSIONS

The EPS developed for the AMALIA lunar rover has been presented, analyzing the system configuration and the main design issues.

Feasibility of single-cell batteries for space vehicles with power in the some-tens of Watt range was demonstrated. A BCDR for this configuration, consisting of a non-insulated high-voltage-gain bidirectional converter with the TLNPC topology was developed and tested, observing remarkable levels of conversion efficiency.

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BIOGRAPHY



Michele Macellari *received the Laurea degree in electronic engineering in 2002 from the University of Perugia, Italy and the Ph.D. in Aerospace Engineering from the Sapienza University in 2006. His major field of interest includes on-board power generation, regulation and distribution systems for aerospace applications. His current research activities include digital control techniques and high efficiency driving for switching power converters.*



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